AREPALLI RAVI KIRAN

arepalliravi678@gmail.com

Mobile: 9542020567

OBJECTIVE:

Enthusiastic and motivated Physical Design fresher seeking an opportunity to apply academic knowledge and skills in a dynamic work environment. Eager to contribute to cutting-edge projects and develop expertise in the field of Physical Design.

TECHNICAL COURSE:

Physical Design Training @SUMEDHADESIGN SystemPvt Ltd, Hyderabad, From October 2024 to May 2025 training in ASIC Physicaldesign, block level implementation and multiple macro placement in TSMC 28nm. Hands-on-experience in building floor planning, placement, congestion analysis, timing closure, clock tree synthesis and routing.

TECHNICAL SKILLSET:

Languages: TCL/TK

Software Tools : Synopsys :dc_shell,icc2_shell,pt_shell.

Platforms: Windows XP/7/8/10/11 and Linux

Other: Microsoft Office

RESPONSIBILITIES:

- Imported Verilog Netlist, real SDC, QrcTech Files and Linked Physical Libraries Performed Sanity Checks
- Floor Planning: used source file for I/O port Locations and boundary. Provided core to I/O Boundary Placed Macros and Fixed Macros and I/O Ports. Deliver PG connection, created Power Straps for Nets VDD, VSS.
- Placement: Add end cap cells, tap cells and set attribute fixed true. inserted buffers for all I/O ports. created placement blockages (Hard, Soft), Create placement and legalized Placement and verified cell density, pin density, Module placement. placement utilization, congestion Report.
- CTS: Define routing rules width and spacing (Layers and width), applied routing rule to clock nets specified CTS Buffers used for optimization
- Routing: Created path groups, define routing layers for routing, set timing driven true for routing.
- Time analysis and optimizing techniques
- DRC and LVS : cleared the open and shorts.

PROJECTS:

Block 1:

Technology	28 nm/9
Macros / Instance	22/218k
Frequency	500 MH
Tools	Design
No.of clocks	1

Challenges:

- Sanity checks performed after synthesis.
- Design constraints like Input and Output delays, uncertainty.
- Tried different Floor plans to overcome the congestion;Optimizing the design with respect to performance.

Block 2:

Technology	TSMC 28nm/9 Layers.
Macros/Instances	6/61K
Max Frequency	500 MHz
No. of clocks	1
Design Tools	ICC2_shell.DC_shell
Responsibilities	Block level netlist to GDSII implementation, which includes Floorplan, Power planning, Placement, CTS, Routing. Also Performed Timing Analysis,Timing closure, ECO's.

Challenges:

- Block was timing critical, created many path groups to meet timing
- Congestion due to high pin density
- Fixed DRVs – max_tran, mac_cap & max_fanout violations

Block 3:

Tools	: dc_shell, icc2_shell,pt_shell.
Description	: Block Level Implementation
Technology Mode	: 28nm TSMC
Instances count	: 450
Frequency	: 400MHZ

Challenges:

- Implemented block design from netlist to GDSII.
- Involved in Floor planning with 75% utilization, Power Plan, Placement, CTS, Routing, and Static timing analysis. Congestion analysis, Timing closure, DRC fixes.

ECO: Setup fixes by route changing in net dominant corners, Timing critical paths fixes by skewing, Custom buffering for long route paths, skewing at common point for multiple paths to reduce hold buffering, SVT to LVT swapping for setup hold conflict paths, Caliber fixes by swapping, size cell and adding buffer on long net length

EDUCATION:

Degree	Institute	Year of Passing	Percentage/CGPA
B.Tech[ECE]	Eluru College of Engineering &Technology	2024	7.08
Intermediate[MPC]	NRI Junior College	2020	6.0
SSC	Manasa E.M High School	2018	8.0

DECLARATION:

I hereby declare that all the above information given by me is true to my knowledge and I hold the responsibility for the above-mentioned information.

Place :

Date :

A.RAVI KIRAN